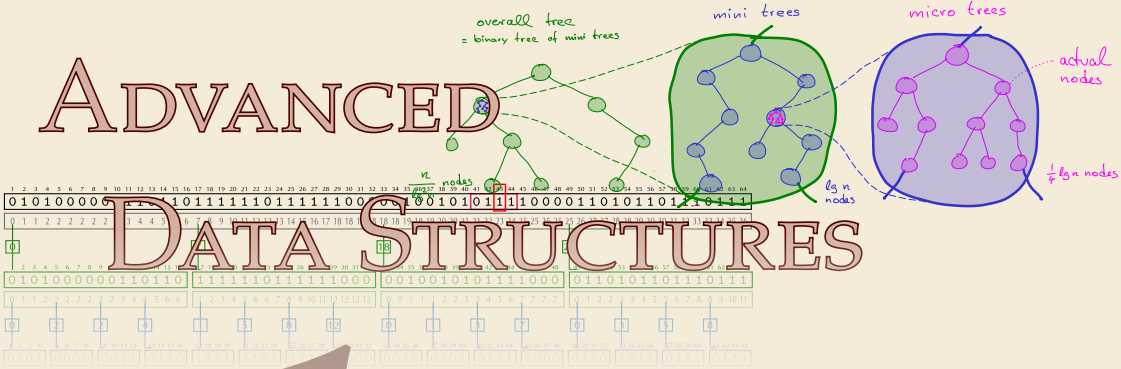
Advanced Data Structures



1 Randomized Trees

Advanced Data Structures · Summer 2026

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CS650 (Summer 2026)
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1 Randomized Trees

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1.2 What's wrong with BBSTs?
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1.12 Fringe Balancing

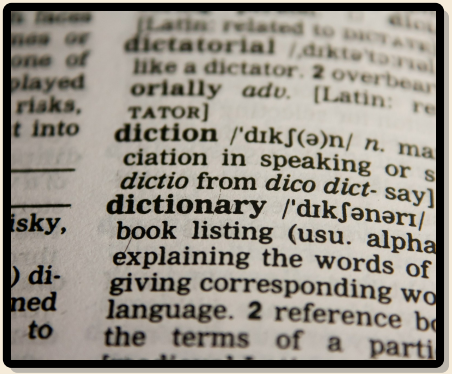
1.1 Recap: Sorted Dictionary and BSTs

Symbol table ADT

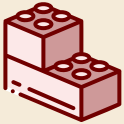
Java: `java.util.Map<K,V>`

Symbol table / Dictionary / Map / Associative array / key-value store:

Python dict `{k:v}`



- `put(`k`,`v`)` *Python* dict: `d[`k`] = `v
  Put key-value pair (k, v) into table
- `get(`k`)` *Python* dict: `d[`k`]`
  Return value associated with key k
- `delete(`k`)` *Python* dict: `del d[`k`]`
  Remove key k (any associated value) form table
- `contains(`k`)` *Python* dict: k `in d`
  Returns whether the table has a value for key k
- `isEmpty(), size()`
- `create()`

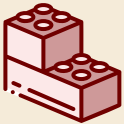


Most fundamental building block in computer science.

(Every programming library has a symbol table implementation.)

Ordered symbol tables

- `min()`, `max()`
  Return the smallest resp. largest key in the ST
- `floor(`x`)`, $\lfloor x \rfloor = \mathbb{Z}.$`floor(`$x$`)`
  Return largest key k in ST with $k \leq x$.
- `ceiling(`x`)`
  Return smallest key k in ST with $k \geq x$.
- `rank(`x`)`
  Return the number of keys k in ST $k < x$.
- `select(`i`)`
  Return the ith smallest key in ST (zero-based, i. e., $i \in [0..n)$)



With select, we can simulate access as in a truly dynamic array!

(Might not need any keys at all then!)

Binary search trees

Binary search trees (BSTs) $\approx$ dynamic sorted array

- binary tree
  - Each node has left and right child
  - Either can be empty (`null`)
- Keys satisfy ***search-tree property***

$$\boxed{\text{all keys in left subtree} \leq \text{root key} \leq \text{all keys in right subtree}}$$

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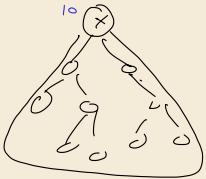
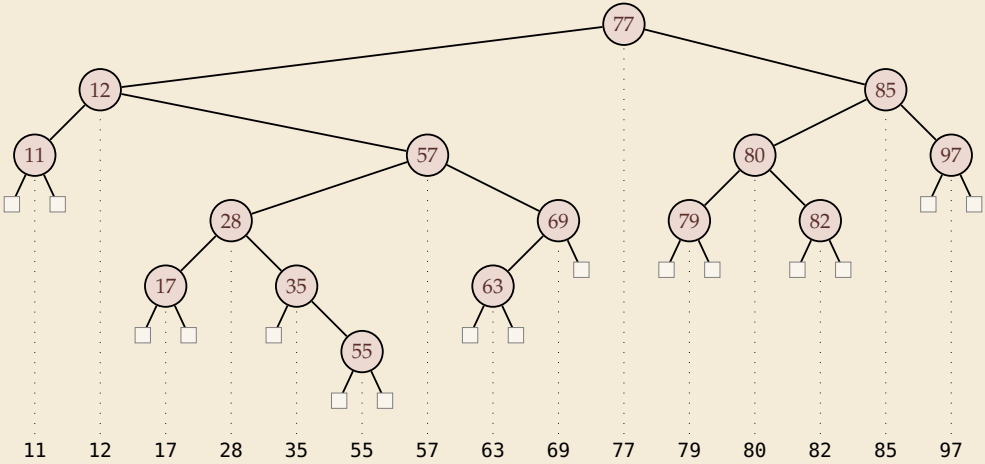
- Standard trick: ***Augmented BSTs***
  - Each node stores the **size** of its **subtree**
  - Easy to maintain upon updates
  - $\rightsquigarrow$ Allows to answer all ordered-symbol-table operations

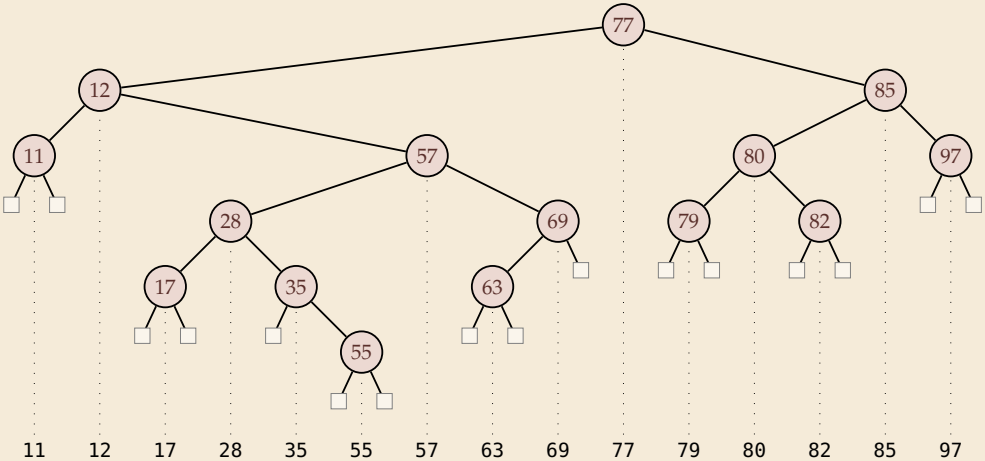


BST example & find



BST insert

Example: Insert 88



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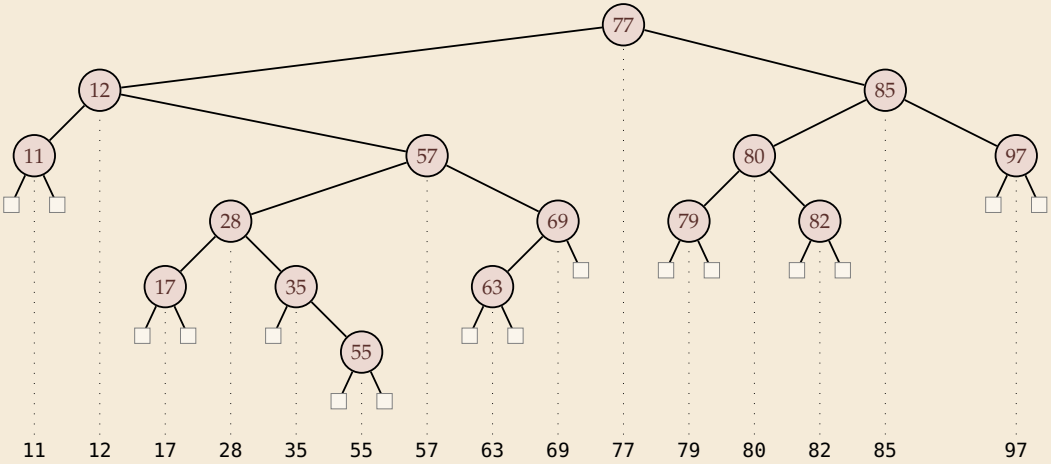

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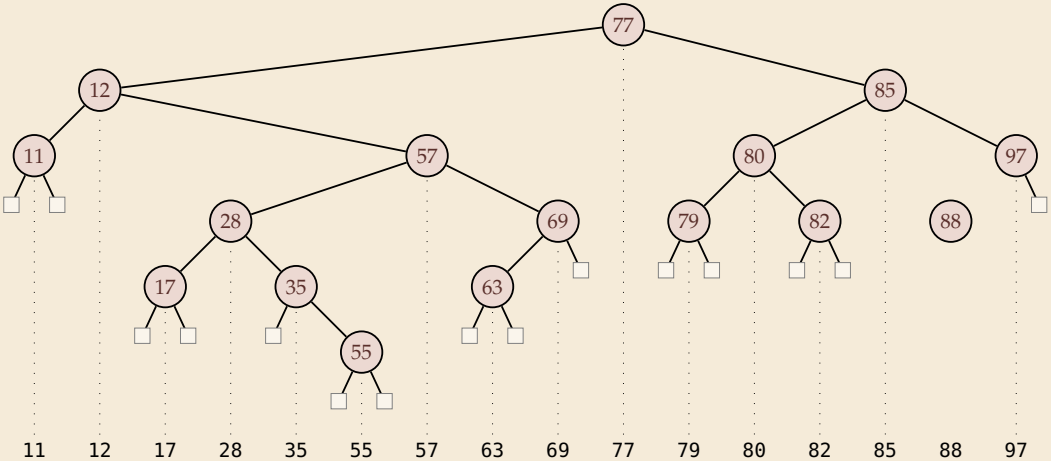

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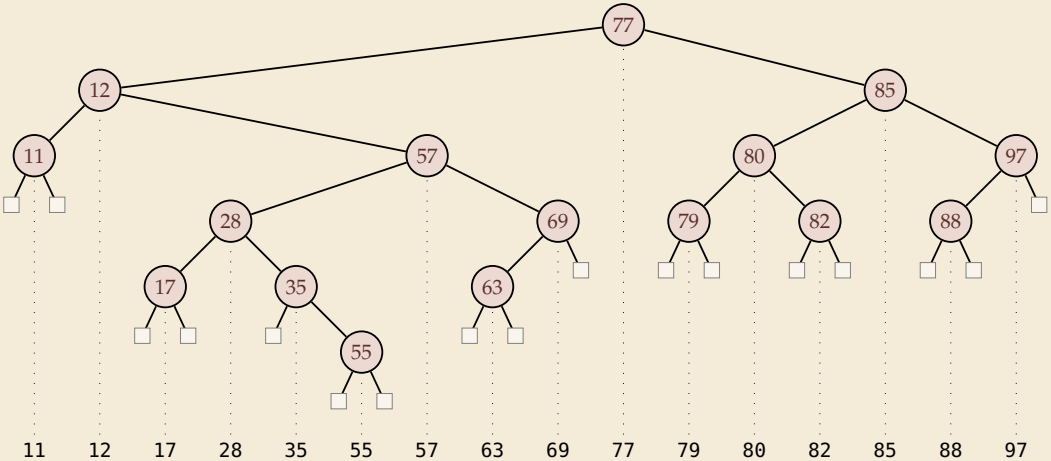
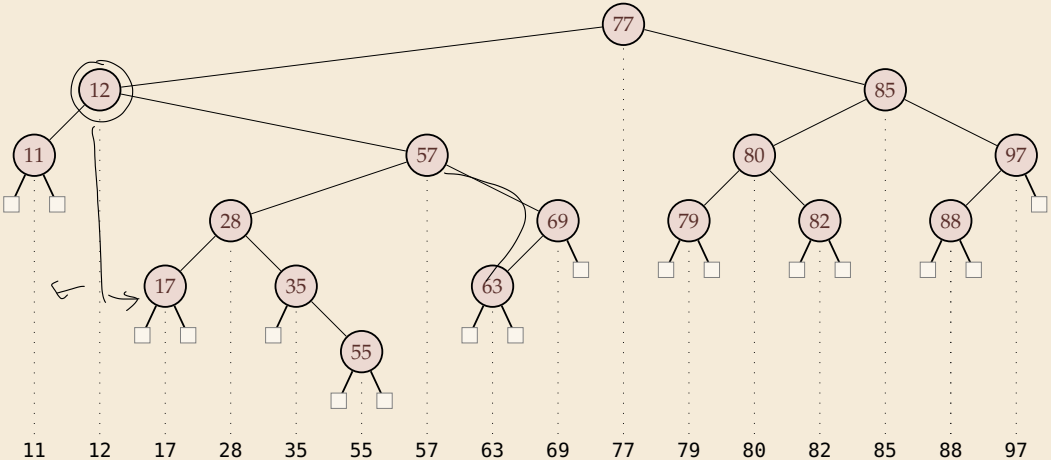


BST delete

Hibbard's deletion

- Easy case: remove leaf, e. g., 11 ⇝ replace by `null`
- Medium case: remove unary, e. g., 69 ⇝ replace by unique child
- Hard case: remove binary, e. g., 85 ⇝ swap with predecessor, recurse



Unbalanced Binary Search Trees

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(n\boldsymbol{h})$
`put(`k`,`v`)`	$O(\boldsymbol{h})$
`get(`k`)`	$O(\boldsymbol{h})$
`delete(`k`)`	$O(\boldsymbol{h})$
`contains(`k`)`	$O(\boldsymbol{h})$
`isEmpty()`	$O(1)$
`size()`	$O(1)$
`min(),max()`	$O(1)$ (if stored)
`floor(`x`),ceiling(`x`)`	$O(\boldsymbol{h})$
`rank(`x`),select(`i`)`	$O(\boldsymbol{h})$

$h =$ *height* of the BST

$1 - 2^{-n}$

- Height h of unbalanced BST depends on insertion order
- satisfies $\lg n \leq h \leq n$

$\lg = \log_2$

$\text{Prob} = 1 - o(1)$

- For **random** insertions, $h = O(\log n)$ in expectation and with high probability

Balanced Binary Search Tree

Balanced binary search trees (BBSTs):

- imposes shape invariant that guarantees $O(\log n)$ height
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  - *red-black trees*

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  - *weight-balanced trees* (BB[α] trees)

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- **Properties**
  - All of them guarantee $h = O(\log n)$ at all times (constants differ!)
  - All of them work with *rotations* along the search path

  $\rightsquigarrow$ $O(\log n)$ extra cost for maintaining balance
  - Further guarantees specific to rule known

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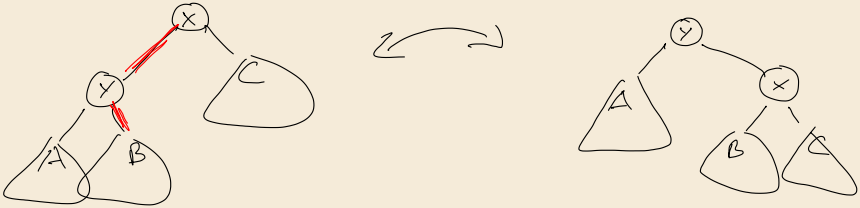
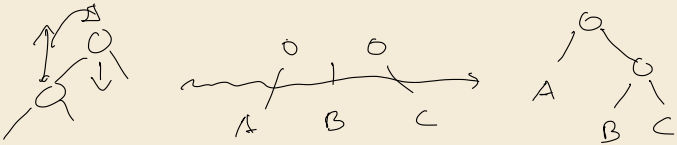
OPEN: *What is the exact average cost (constant in front of $\lg n$) of insertion in AVL trees / red-black trees?*

A Primitive for BSTs: Rotations





1.2 What's wrong with BBSTs?

Red-Black Tree Implementation

`RedBlackTree` from Sedgewick & Wayne (excerpt) `algs4.cs.princeton.edu/code`

- showing only the bare-bones core: `put` (insert) and `delete`
- no rank-based operations
- full class has 750 lines (and this is a good and compact implementation!)

```
public class RedBlackBST<Key extends Comparable<Key>, Value> {
    public void put(Key key, Value val) {
        root = put(root, key, val);
        root.color = BLACK;
    }

    private Node put(Node h, Key key, Value val) {
        if (h == null) return new Node(key, val, RED, 1);

        int cmp = key.compareTo(h.key);
        if      (cmp < 0) h.left  = put(h.left,  key, val);
        else if (cmp > 0) h.right = put(h.right, key, val);
        else              h.val   = val;

        // fix-up any right-leaning links
        if (isRed(h.right) && !isRed(h.left))      h = rotateLeft(h);
        if (isRed(h.left)  &&  isRed(h.left.left)) h = rotateRight(h);
        if (isRed(h.left)  &&  isRed(h.right))     flipColors(h);
        h.size = size(h.left) + size(h.right) + 1;

        return h;
    }

    private Node deleteMin(Node h) {
        if (h.left == null)
            return null;

        if (!isRed(h.left) && !isRed(h.left.left))
            h = moveRedLeft(h);

        h.left = deleteMin(h.left);
        return balance(h);
    }
```

```
    private Node deleteMax(Node h) {
        if (isRed(h.left))
            h = rotateRight(h);

        if (h.right == null)
            return null;

        if (!isRed(h.right) && !isRed(h.right.left))
            h = moveRedRight(h);

        h.right = deleteMax(h.right);

        return balance(h);
    }

    private Node delete(Node h, Key key) {
        // assert get(h, key) != null;

        if (key.compareTo(h.key) < 0)  {
            if (!isRed(h.left) && !isRed(h.left.left))
                h = moveRedLeft(h);
            h.left = delete(h.left, key);
        }
        else {
            if (isRed(h.left))
                h = rotateRight(h);
            if (key.compareTo(h.key) == 0 && (h.right == null))
                return null;
            if (!isRed(h.right) && !isRed(h.right.left))
                h = moveRedRight(h);
            if (key.compareTo(h.key) == 0) {
                Node x = min(h.right);
                h.key = x.key;
```

```
                h.val = x.val;
                // h.val = get(h.right, min(h.right).key);
                // h.key = min(h.right).key;
                h.right = deleteMin(h.right);
            }
            else h.right = delete(h.right, key);
        }
        return balance(h);
    }

    private Node rotateRight(Node h) {
        assert (h != null) && isRed(h.left);
        Node x = h.left;
        h.left = x.right;
        x.right = h;
        x.color = h.color;
        h.color = RED;
        x.size = h.size;
        h.size = size(h.left) + size(h.right) + 1;
        return x;
    }

    private Node rotateLeft(Node h) {
        assert (h != null) && isRed(h.right);
        Node x = h.right;
        h.right = x.left;
        x.left = h;
        x.color = h.color;
        h.color = RED;
        x.size = h.size;
        h.size = size(h.left) + size(h.right) + 1;
        return x;
    }
```

```
    private void flipColors(Node h) {
        h.color = !h.color;
        h.left.color = !h.left.color;
        h.right.color = !h.right.color;
    }

    private Node moveRedLeft(Node h) {
        flipColors(h);
        if (isRed(h.right.left)) {
            h.right = rotateRight(h.right);
            h = rotateLeft(h);
            flipColors(h);
        }
        return h;
    }

    private Node moveRedRight(Node h) {
        flipColors(h);
        if (isRed(h.left.left)) {
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            flipColors(h);
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        return h;
    }

    private Node balance(Node h) {
        if (isRed(h.right) && !isRed(h.left))    h = rotateLeft(h);
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Can we have something simpler?

In this and the next unit, we will look into alternatives to avoid lengthy case distinctions.

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Disclaimer: It seems that something has to give.

All known (much) more elegant BST alternatives are either *randomized* or *amortized*.

- However, this is often tolerable.
- Complexity of code can also come with running-time penalty;
  if we avoid that, might be an overall speedup.
  (Typically not the case for well-tested library implementations, though.)

Recall: Average-Case Analysis vs. Randomized Algorithms

This unit: *Randomized solutions*.

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- algorithm is **deterministic**
  same input, same computation

Randomized Algorithm (here)

- algorithm is **not** deterministic
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Confusingly enough, the analysis (technique) is often the same!

But: Implications are quite different; randomization is much more versatile and robust.

1.3 Random BSTs

Random BSTs are good (enough)

*Let's first do some wishful thinking: assume we're dealing with **random** inputs only.*

- ⇝ Let's first see if we like the performance in this case.
- If so, will try and see how to **enforce** this randomness later.

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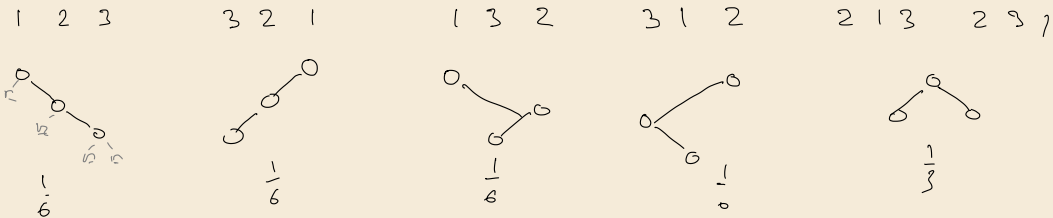
Random here means:

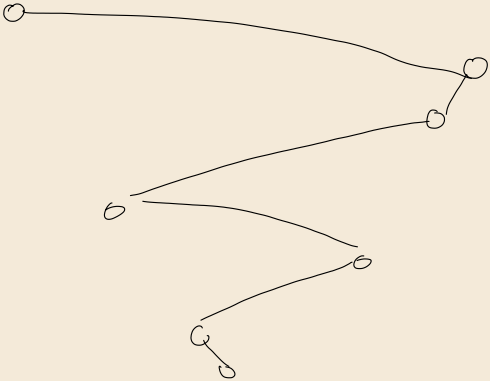
Definition 1.1 (Random BST)

A ***Random BST*** of size n is the (random) shape of an initially empty BST after successively inserting a random permutation of $[n]$.

$= \{1, \dots, n\}$

Example: $n = 3$





$\approx 2^n$ maximally unbalanced trees

#binary trees on n nodes

nth Catalan number

$$= \binom{2n}{n} \cdot \frac{1}{n+1}$$

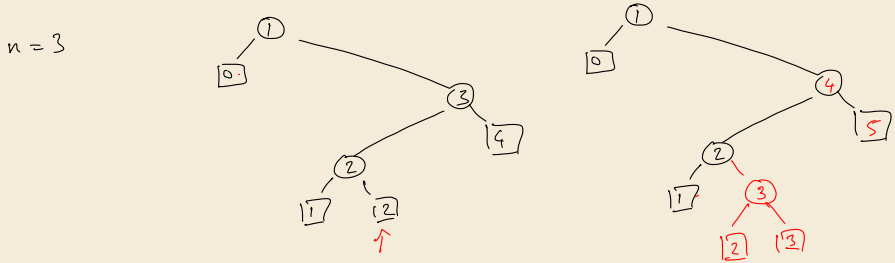
$$\approx 4^n / n^{3/2}$$

$$n! \approx \left(\frac{n}{e}\right)^n$$

Iterative randomness

Lemma 1.2 (Random insertion yields random BST)

Let $n \geq 0$ and T_n a random BST over n keys. Inserting an element equally likely in one of the $n+1$ *gaps* in T_n (external leaves) results in a new BST T_{n+1} that has the same shape as a random BST on $n+1$ keys.



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Proof:

Insertion order for T_{n+1} is a permutation of $[n+1] = \{1, \ldots, n+1\}$, $[0..n]$ which is in bijection with a permutation of $[n]$ and a *insertion point (leaf)* in $[0..n]$, where we increment all values right of the insertion leaf by 1.

$= [1..n+1]$

$= \{1, 2, \ldots, n+1\}$

Example: $1, 2, 4, 7, 6, 5 \mid 3 \;\widehat{=}\; ((1, 2, 3, 6, 5, 4), 2)$

Recall: Events

something we can assign a probability to

$A \in \mathcal{F}$ is called an *event* of probability space $(\Omega, \mathcal{F}, \mathbb{P})$; also a *measurable set*.

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Basic properties

- $\mathbb{P}\left[\overline{A}\right] = 1 - \mathbb{P}[A]$ counter-probability $\quad (\overline{A} = \Omega \setminus A)$
- $\mathbb{P}\left[\bigcup A_i\right] \leq \sum_i \mathbb{P}[A]$ the *union bound* (a.k.a. Boole's inequality a.k.a. σ-subadditivity)

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- $\{A_1, \ldots, A_k\}$ *(mutually) independent* $\iff \mathbb{P}\left[\bigcap_i A_i\right] = \prod_i \mathbb{P}[A_i]$

  An infinite set of events is mutually independent if every finite subset is so.

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discrete, $\mathcal{F} = 2^{\Omega}$ (powerset of Ω)

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  An infinite set of events is mutually independent if every finite subset is so.

  k-wise independence means that only all size-k subsets are independent.
- *conditional probability* for A given B: $\mathbb{P}[A \mid B] = \mathbb{P}[A \cap B] / \mathbb{P}[B]$
  generally undefined if $\mathbb{P}[B] = 0$
- *law of total probability*: If $\Omega = B_1 \dot\cup B_2 \dot\cup \cdots$ is a partition of Ω, we have

$$\mathbb{P}[A] = \sum_{\substack{i \\ \mathbb{P}[B_i] \neq 0}} \mathbb{P}[A \mid B_i] \cdot \mathbb{P}[B_i].$$

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Random variables (r.v.) $X : \Omega \to \mathcal{X}$; often $\mathcal{X} = \mathbb{R}$ (in general spaces: only *measurable* functions)

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Basic properties and conventions:

- event $\{X = x\}$ is defined as $\{\omega \in \Omega : X(\omega) = x\}$.
- For event A define the indicator r.v. $\mathbb{1}_A$ via $\mathbb{1}_A(\omega) = [\omega \in A] = \begin{cases} 1 & \omega \in A \\ 0 & \text{else} \end{cases}$

Recall: Random Variables

Random variables (r.v.) $X : \Omega \to \mathcal{X}$; often $\mathcal{X} = \mathbb{R}$ (in general spaces: only *measurable* functions)

Basic properties and conventions:

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Independence:

- X and Y independent $\iff \mathbb{P}[X = x \wedge Y = y] = \mathbb{P}[X = x] \cdot \mathbb{P}[Y = y]$ for all x, y.

  (Naturally follows from independent events)

- a sequence of *i.i.d.* r.v. $X_1, X_2, \ldots$ (*independent and identically distributed*) has $X_i \overset{\mathcal{D}}{=} X_1$ and $\{X_i\}_{i \geq 1}$ are mutually independent
  - typical example: sequence of coin tosses (with same coin)

Alternative Random Model

Corollary 1.3

A BST built by inserting n i.i.d. Uniform$(0, 1)$ r.v. has the shape of a random BST.

1.4 Properties of Random BSTs

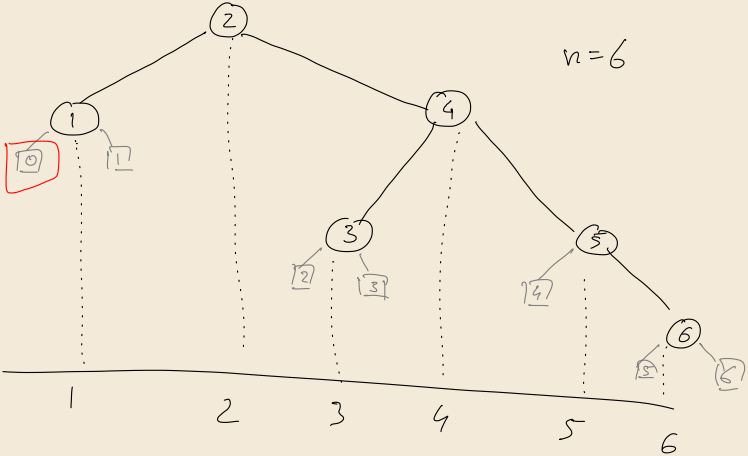


Figure 6.16 A binary search tree built from 237 randomly ordered keys

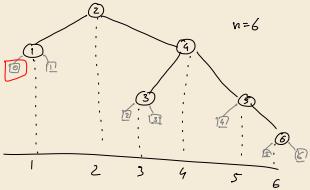
Analysis of Random BSTs I

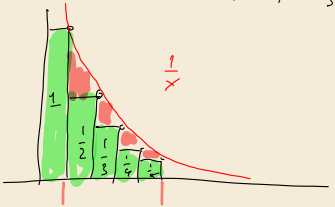
$$\frac{H_n}{\ln n} \to 1 \quad (n \to \infty)$$

Theorem 1.4 (Expected depth of leftmost leaf)

The *expected depth* (number of edges from root) of the leftmost external leaf (leaf for $-\infty$) in a random BST on $n \geq 1$ nodes is $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \sim \ln n$.

$$= \sum_{i=1}^{n} \frac{1}{i}$$





$$\int_1^n \frac{1}{x}\,dx \approx \ln(n) - \ln(1)$$

Analysis of Random BSTs I

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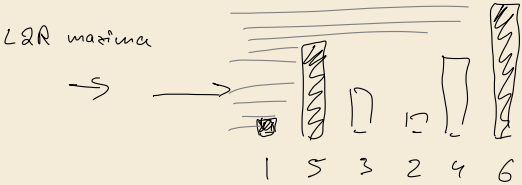
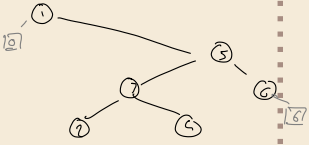
Proof:

$\text{depth}(\boxed{0}) = \#\text{left-to-right minima } \text{L2R}_n \text{ in insertion sequence}$

$$\text{L2R}_n = \sum_{i=1}^{n} X_i \qquad \text{for } X_i = [\text{position } i \text{ is a l2r min}]$$





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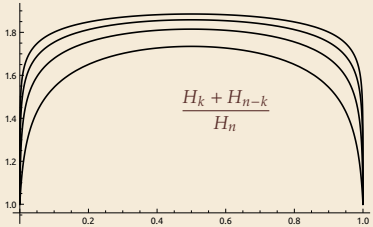
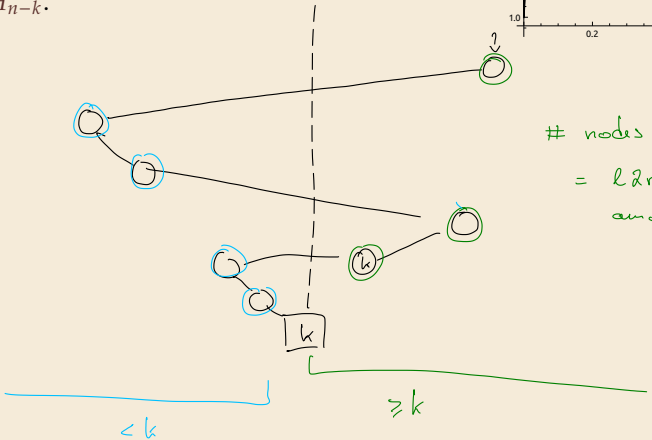
$$\mathbb{E}[\text{L2R}_n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{1}{i} = H_n.$$ ■

Analysis of Random BSTs II

Theorem 1.5 (Expected depth of kth leaf)

The *expected depth* of the *kth external leaf* $\boxed{k}$ (for $k = 0, \ldots, n$) in a random BST on $n \geq 1$ keys is $H_k + H_{n-k}$.

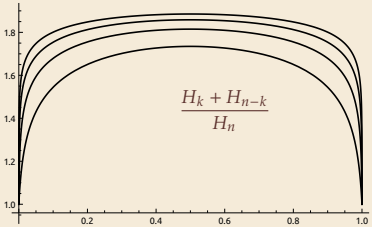




nodes right of k

= l2r minima among keys ≥ k

Analysis of Random BSTs II



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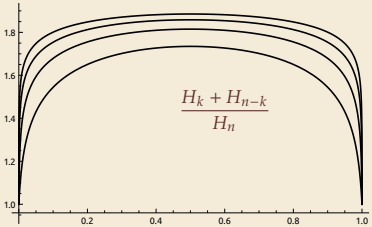
Analysis of Random BSTs II



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Proof:

$k - \frac{1}{2}$

$$\begin{aligned}
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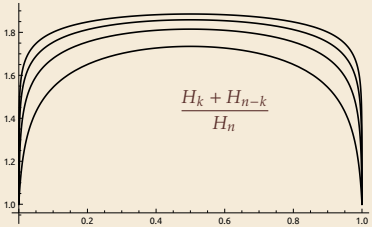
Analysis of Random BSTs II



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&= \#\text{l2}\overset{\text{max}}{\cancel{\text{r-min}}}\text{s among keys} < x \;+\; \#\text{l2}\overset{\text{min}}{\cancel{\text{r-m}}}\text{axs with keys} > x
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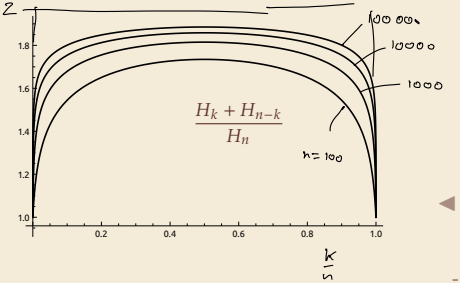
Analysis of Random BSTs II



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&= \text{\#l2r-mins among keys} < x \;+\; \text{\#l2r-maxs with keys} > x
\end{aligned}$$

Since there are k keys $< x$ and $n - k$ keys $> x$, the same derivation as above applies. ■

Analysis of Random BSTs III

$$\mathbb{E}\left[\sum_{k=0}^{n} \text{depth}(\boxed{k})\right] = \sum_{k=0}^{n} (H_k + H_{n-k})$$

$$= 2\sum_{k=0}^{n} H_k$$

Corollary 1.6 (Depth of typical leaf)

Consider a random BST T_n of n keys.

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n,$$

(a) The *expected external path length* of T_n is
$2(n+1)(H_{n+1} - 1) \;=\; 2n \ln n - 2(1-\gamma)n \;\pm\; O(\log n)$. ($\gamma \approx 0.5772$ the *Euler-Mascheroni constant*)

(b) The depth of the αnth leaf in a random BST of n keys $\sim \underline{2 \ln n}$ as $n \to \infty$ for any fixed $\alpha \in (0,1)$.

$1.39 \log_2 n$

$$H_k + H_{n-k} = H_{\alpha n} + H_{(1-\alpha)n}$$

$$\sim \ln(\alpha n) + \ln((1-\alpha)n)$$

$$= 2\ln(n) + \ln(\alpha) + \ln(1-\alpha)$$

$$\sim 2\ln(n)$$

$$H_{n+1} = H_n + \frac{1}{n+1}$$

recursion tree of Quicksort

Detour: When Expectation Isn't Enough

- Two hypothetical algorithms:
  - A takes 1 step in half the cases and 3 steps otherwise
  - B takes 1 step in 99% of cases and **101** steps otherwise
  - $\rightsquigarrow$ both have expected costs of 2 steps.
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  - probably want A ... certainly would want to be able to distinguish them!

- **Goal:** Strengthen algorithms so $time(x)$ rarely far from $\mathbb{E}[time(x)]$
  - formally: bound probability that X (far) exceeds $\mathbb{E}[X]$
    - $\rightsquigarrow$ *concentration bounds* a.k.a. *tail inequalities*
  - 👍 can then compare these typical times again
  - 👍 also obtain more reliable algorithms
  - $\rightsquigarrow$ Let's establish some tools for that!

Detour: With High Probability

Definition 1.7 (With high probability)

We say

- an event $X = X(n)$ happens *with high probability (w.h.p.)* when $\forall c \;:\; \mathbb{P}[X(n)] \stackrel{!}{=} 1 \pm O(n^{-c})$ as $n \to \infty$. $1 - d(c) \cdot n^{-c}$
- a random variable $X = X(n)$ is *in $O(f(n))$ with high probability (w.h.p.)* when $\forall c \; \exists d \;:\; \mathbb{P}[X \leq d f(n)] = 1 \pm O(n^{-c})$ as $n \to \infty$.
  (This means, the constant in $O(f(n))$ may depend on c.)

Height / Depth of a leaf in Random BST $H = O(\log n)$ w.h.p.

$c = 10 \rightsquigarrow d_1, d_2 : \mathbb{P}[H \geq d_1 \lg n] \leq d_2 n^{-10}$

(for example, $\mathbb{P}[\text{height} \geq 42 \ln n] \leq 2 n^{-7.4}$)

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  (This means, the constant in $O(f(n))$ may depend on c.) ◀
- Very strong notion: failure probability smaller than any polynomial

$\rightsquigarrow$ If A succeeds w.h.p. then also polynomially many repetitions of A succeed w.h.p.

Lemma 1.8 (Repetitions w.h.p.)

Suppose A is an algorithm that w.h.p. does not fail.
In n^d *independent* repetitions of A on inputs of size n, w.h.p. *no* repetition fails. ◀

Detour: With High Probability [2]

Proof (Lemma 1.8):

Let c from the definition of w.h.p. be given.

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Detour: Chernoff Bounds

Strong concentration inequalities require assumptions on distribution of X.
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Theorem 1.9 (Chernoff Bound for Bernoulli trials)

Let $X_1, \ldots, X_n \in \{0,1\}$ be *(mutually) independent* with $X_i \stackrel{\mathcal{D}}{=} \mathrm{B}(p_i)$.
Define $X = X_1 + \cdots + X_n$ and $\mu = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = p_1 + \cdots + p_n$. Then holds

$$\forall \delta > 0 \quad : \quad \mathbb{P}[X \geq (1+\delta)\mu] \;<\; \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

$$\forall \delta \in (0,1] \quad : \quad \mathbb{P}[X \geq (1+\delta)\mu] \;\leq\; \underline{\exp(-\mu\delta^2/3)}$$

Proof (Sketch):

Apply Markov's Inequality to $Y = e^{tX}$, then choose convenient t.

(more details in *Advanced Algorithms*)

Detour: Chernoff Bound for Binomial Distribution

The algorithmically most widely used special case has identical coin flips.

Corollary 1.10 (Chernoff Bound for Binomial Distribution)

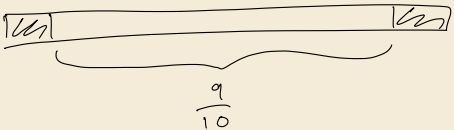
Let $X \stackrel{\mathcal{D}}{=} \mathrm{Bin}(n,p)$. Then

$$\forall \delta \geq 0 \quad : \quad \mathbb{P}\left[\left|\frac{X}{n} - p\right| \geq \delta\right] \leq 2\exp(-\overbrace{2\delta^2 n}^{\to\infty})$$

$$\delta > \frac{1}{\sqrt{n}} = n^{-0.5} \qquad n^{-0.499}.$$

$\rightsquigarrow$ $\boxed{\mathrm{Bin}(n,p) \in np \pm n^{0.501} \text{ w.h.p.}}$

$p = 0.1$



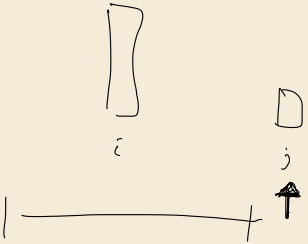
Analysis of Random BSTs – Concentration

Theorem 1.11 (Concentration of left-to-right minima)

The number of left-to-right minima in a permutation of length n is in $O(\log n)$ w.h.p.
Hence, the above expected results hold with high probability (up to constant factors).

Proof (Idea):
Via *inversion table* of permutation:
$a_1, \ldots, a_n$ permutation of $[n]$ in bijection with $b_1, \ldots, b_n$ where

$$\begin{aligned} b_j &= \#\text{inversions of form } (\bullet, j) \\ &= \#\text{values left of } \cancel{i}\ j \text{ that are} > \cancel{i}\ j \end{aligned}$$



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*greater and left of *number* j*

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$$\text{L2RMax}(a) = \sum_{j=1}^{n} [b_j = 0] \rightsquigarrow \text{a sum of independent Bernoulli trials.}$$

+ symmetry

Hook-Length Formula for Random BSTs

For a given tree, the probability to see this shape can be computed recursively over the tree structure:

Theorem 1.12 (Random BST Distribution)

Let T_n be a binary tree. The probability that Random BSTs attains the shape T_n after n insertions is

$$\Pr[T_n] = \begin{cases} 1 & n = 0, \\ \frac{1}{n} \cdot \Pr[T_L] \cdot \Pr[T_R] & n \geq 1, \end{cases}$$ (for T_L and T_R the left resp. right subtrees of T).

randomness preservation! filtering out values $\leq x$

again a random permutation

$\mathbb{P}\left[\begin{array{c} \circ \\ \circ \quad \circ \end{array}\right] = \frac{1}{3}$ $\qquad \frac{1}{3}$

$\mathbb{P}\left[\begin{array{c} \circ \\ \circ \end{array}\right] = \frac{1}{3} \cdot \mathbb{P}\left[\circ\right] = \frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{6}$

More AofA (Analysis of Algorithms)

Remark 1.13 (Knowledge on Random BSTs)

Random BSTs are extremely well-studied in Analysis of Algorithms.
A few more results (all proven in the literature):

(a) The **expected height** is $\alpha \ln n - \beta \ln \ln n \pm O(1)$ with $\alpha \approx 4.311$ and $\beta \approx 1.953$.

(b) The **height** divided by $\ln n$ **converges in probability** to the constant α.

(c) The number X_{nk} of **external leaves at depth** k satisfies $\mathbb{E}[X_{nk}] = \frac{2^k}{n!}\left[{n \atop k}\right]$.

(d) The **depth** of a typical **leaf** divided by $\ln n$ **converges in probability** to 2.

(e) The standardized **depth** of a random leaf **converges** in distribution to a standard **normal distribution**.

(f) The same is true for the standardized depth of a random internal node.

(g) Let D_n be the **depth of the nth inserted node**. Then $(D_n - \ln n)/\sqrt{\ln n}$ converges in distribution to a standard **normal distribution**. ◀

⇝ In many ways, random BSTs are close to perfectly balanced.

⇝ If only we can get the performance of random BSTs, we're happy.

Caveat: Random Deletions ≠ Random BSTs!

⇝ *Unbalanced BSTs have **great** performance **if** insertions come in random order.*

Interesting fact: *no longer true* if there are also *deletions*!

After long sequence of random inserts and deletes: expected height $\Theta(\sqrt{n})$, not $\Theta(\log n)$ (!)

Reason: Hibbard's deletion algorithm destroys randomness!

Animations: `http://algs4.cs.princeton.edu/32bst`

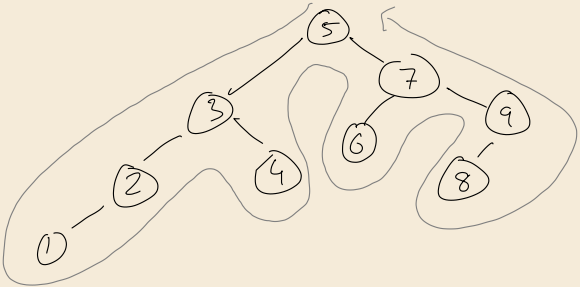
1.5 Treaps

Treaps

Observation: The *preorder* sequence of the keys fully determines a BST since

- each BST has unique preorder, and
- each preorder generates a unique BST by inserting keys in preorder into an initially empty tree.

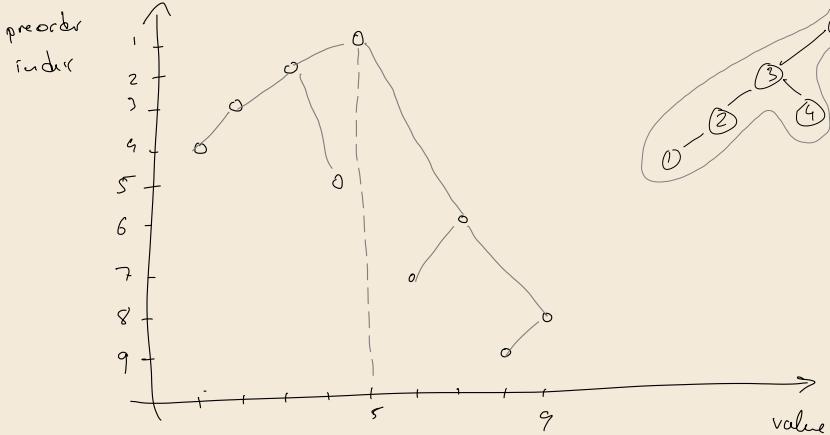
⇝ Enforcing the preorder corresponding to a **random** BST suffices!
. . . but how? We have no control over the insertion of keys!

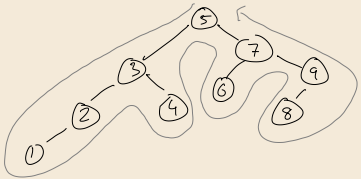


5 3 2 1 4 7 6 9 8
1 2 3 4 5 6 7 8 9

5	3	2	1	4	7	6	9	8
1	2	3	4	5	6	7	8	9

Treap = Cartesian tree





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$\rightsquigarrow$ Enforcing the preorder corresponding to a **random** BST suffices!
. . . but how? We have no control over the insertion of keys!

Idea: Separate *key values* from *rank in insertion order* using random **"priorities"**.

Definition 1.14 (Treap)

Let $S = \{(k_1, p_1), \ldots, (k_n, p_n)\}$ be a set of *key-priority pairs* where $k_i \in K$ and $p_i \in [0, 1]$ for K some totally ordered universe.
A *treap* for S is a binary tree with n internal nodes labeled by the key-priority pairs so that

(a) the *search tree property* holds w. r. t. the keys, and

(b) the *heap property* holds w. r. t. the priorities.

There can be only one

Theorem 1.15 (Treaps are unique)

Let S be a set of n key-priority pairs where all keys and all priorities are distinct. Then there is *exactly one treap* for S.

Proof:

existence: use algorithm w/ sliding down horizontal line

uniqueness by induction

IB $n = 1$ ✓

IS: find $i = \arg\min p_j$ k_i = root

$\Rightarrow$ treaps for all (k_j, p_j) with $k_j < k_i$

resp. (k_j, p_j) " $>$

are unique

Randomized Treaps

Definition 1.16 (Randomized Treaps)

A *randomized treap* is the unique treap that results from given keys $k_1, k_2, \ldots$ where (upon insertion) we assign k_i a priority $p_i \overset{\mathcal{D}}{=} \text{Uniform}(0, 1)$ independent of all previous priorities.

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The (random) shape of a randomized treap for n keys has the *same distribution* as random BST with n keys. ◀

Corollary 1.18 (Search Costs)

All results for random BSTs apply, in particular: // $1.99 \log n$

(a) Expected search costs (#comparisons) $< 2 \ln n + 1$.

(b) Search costs in $O(\log n)$ w.h.p. ◀

1.6 Updates in Treaps

Insertions and Deletions in Randomized Treaps

Up to now: *static* view on treaps.

But can we efficiently turn a randomized treap for $k_1, \ldots, k_n$ into one for $k_1, \ldots, k_{n+1}$? And vice versa?

Insertions and Deletions in Randomized Treaps

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But can we efficiently turn a randomized treap for $k_1, \ldots, k_n$ into one for $k_1, \ldots, k_{n+1}$? And vice versa?

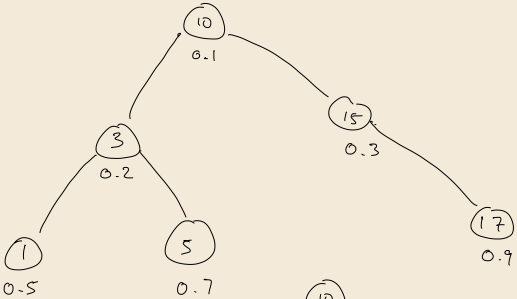
Yes!

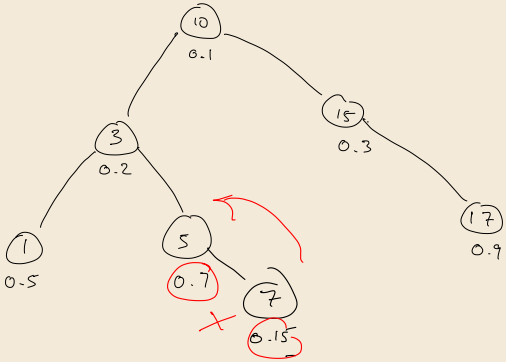
- **Insert:** Start as in plain BST, then *rotate up* until heap property holds.
- **Delete:** Rotate node down (as if priority was $+\infty$) until it is a leaf, then remove it.

Conceptually very simple!

$\rightsquigarrow$ *all operations in $O(\log n)$ time w.h.p.!* in particular # rotations = $O(\log n)$ whp

insert 7


10
0.1
3
0.2
15
0.3
1
0.5
5
0.7
17
0.9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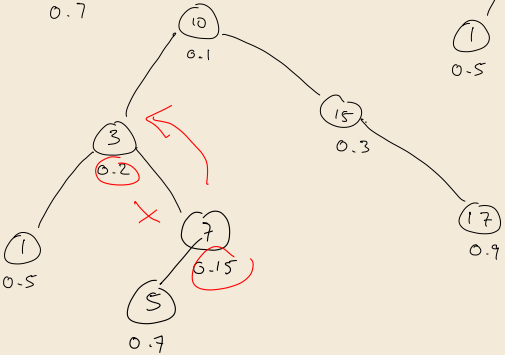
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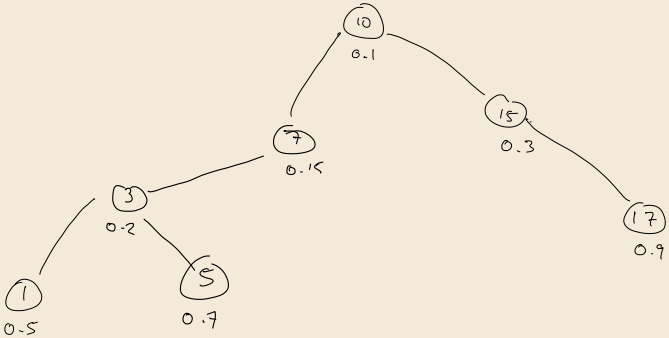


$\notin$ exam : Zip trees $\approx$ treap w/ ties for priorities are allowed / exploited

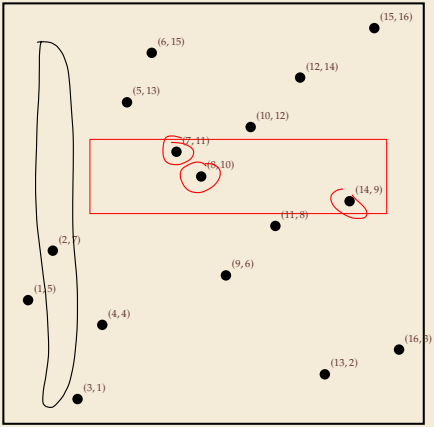
Excursion: Why care about counting rotations?

For secondary data structures, cost of a rotation can be *linear in subtree size*.

Example: *Range trees*

kd-trees, quad trees

2d-orthogonal range search



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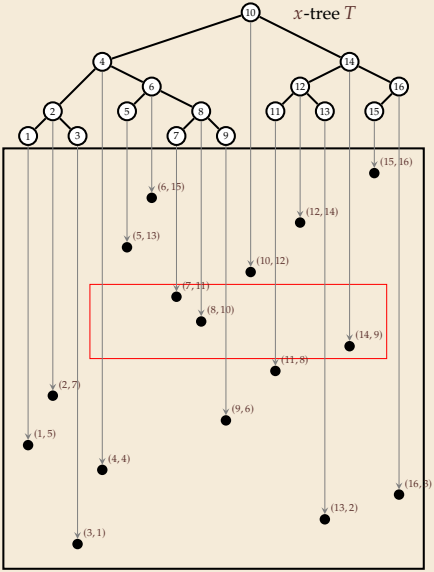

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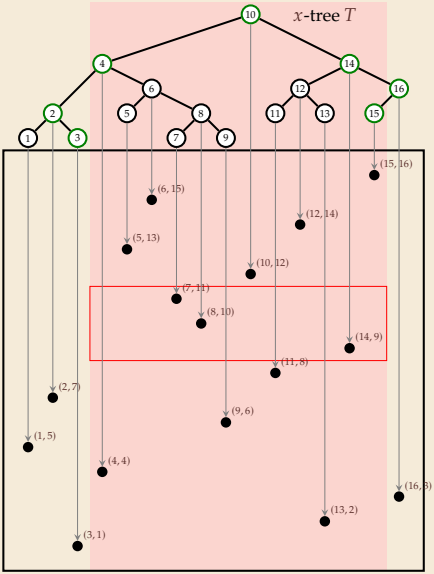

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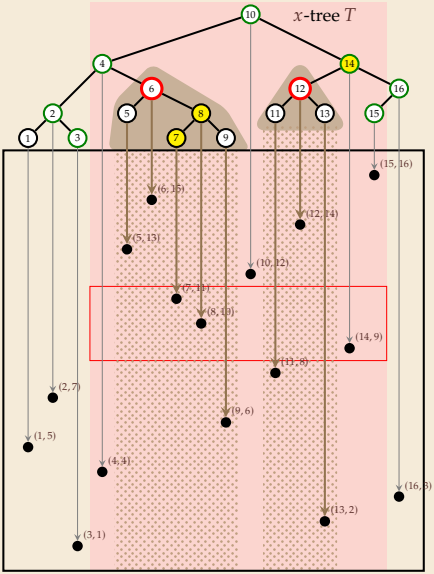

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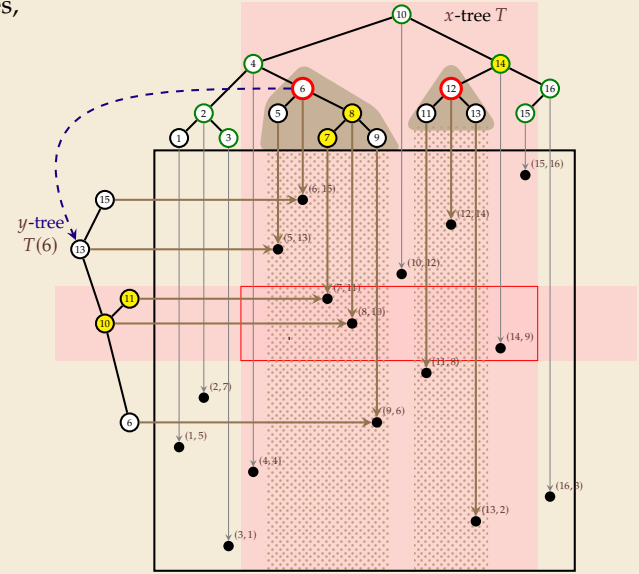
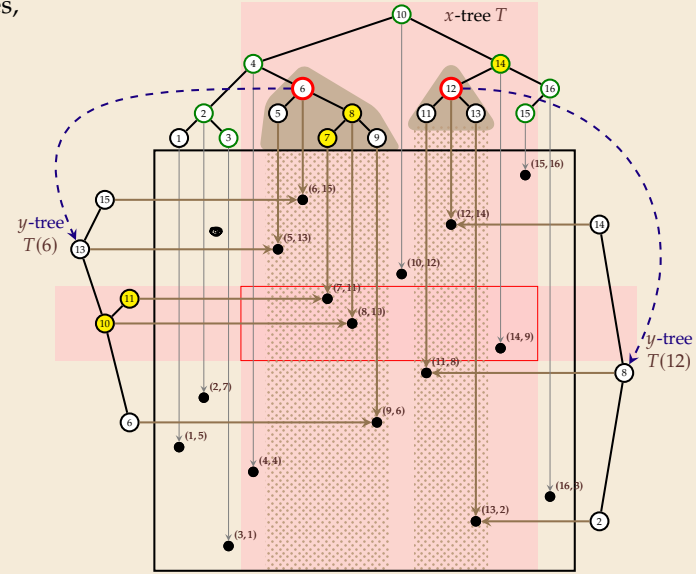


Excursion: Why care about counting rotations?

For secondary data structures, cost of a rotation can be *linear in subtree size*.

Example: *Range trees*

insert (x,y)
needs (potentially)
to rotate in x-treap



Ancestor Indicators

Can actually bound number of rotations much more tightly!

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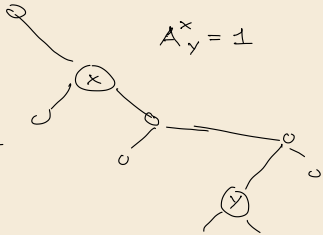
For that, we need closer look at depths of *internal nodes* in random BSTs.
Analysis possible based on handy notion:

Ancestor Indicators



Can actually bound number of rotations much more tightly!

For that, we need closer look at depths of *internal nodes* in random BSTs.
Analysis possible based on handy notion:

Lemma 1.19 (Ancestor indicators)

Let T_n be a random BST with keys $[n]$ and denote by $A^x_y = [x \text{ is a } \textit{proper} \text{ ancestor of } y]$ for $x, y \in [n]$.
(This means $A^x_x = 0$ and for $x \neq y$, $A^x_y = 1$ iff x lies on the path from the root to y.)
Then holds:

(a) $A^x_y = 1$ iff x was the *first* among the keys $[x..y] \cup [y..x]$ that was inserted into T_n.

(b) $A^x_y = 1$ iff x and y are *directly compared* by randomized Quicksort during a partitioning step using pivot x.

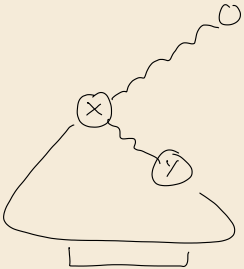
(c) $\Pr[A^x_y = 1] = \Pr[A^y_x = 1] = \dfrac{1}{|y-x|+1}$ for $x \neq y$.

Ancestor Indicators – Proof

(a) $A_y^x = 1$ iff x was the *first* among the keys $[x..y] \cup [y..x]$ that was inserted into T_n.

Proof: $\boxed{x < y \quad \text{wlog}}$

(a) "$\Rightarrow$" $A_y^x = 1$



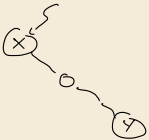
consider keys $[x..y]$

any other key $z \in (x..y]$

$\Rightarrow$ z separates x from y

$\Rightarrow$ x not an ancestor of y $\}$

"$\Leftarrow$" x inserted first among $[x..y]$



(c)

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from (a) $\mathbb{P}[A^*_y = 1] = \mathbb{P}[x \text{ inserted first among } [x..y]]$

$$= \frac{1}{|y-x|+1}$$

$\Rightarrow$ analysis of Quicksort!

never compare same pair twice

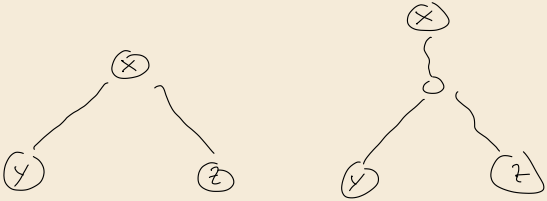
$$C = \sum_{1 \le i < j \le n} \underbrace{[i \text{ compared to } j]}_{\overset{(b)}{=} A_j^i + A_i^j} \overset{(c)}{=} \sum_{1 \le i < j \le n} \frac{2}{j-i+1}$$

Common Ancestor Indicators

Remark 1.20 (Common ancestor indicators)

Idea generalizes to $C^x_{y,z} = [x \text{ is common ancestor of } y \text{ and } z]$: $= A^x_y \cdot A^x_z$

$$\Pr[C^x_{y,z} = 1] \;=\; \frac{1}{\max\{x,y,z\} - \min\{x,y,z\} + 1}.$$



Depth of Internal Nodes

Theorem 1.21 (Expected depth of kth node)

The *expected depth* of the *kth internal node* (for $k = 1, \ldots, n$) in a random BST on $n \geq 1$ nodes is $H_k + H_{n-k+1} - 2$. ◀

Recall: $\mathbb{E}[\text{depth}(\boxed{k})] \;=\; H_k + H_{n-k}$.

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Recall: $\mathbb{E}[\text{depth}(\boxed{k})] \;=\; H_k + H_{n-k}$.

Proof:

$$\text{depth}(k) \;=\; \sum_{x=1}^{n} A_k^x$$

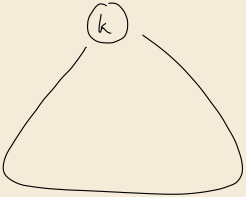
$$\mathbb{E}[\text{depth}(k)] = \sum_{x=1}^{n} \mathbb{P}[A_k^x = 1] = \sum_{x=1}^{k-1} \frac{1}{k-x+1} + \sum_{x=k+1}^{n} \frac{1}{x-k+1}$$

$$= \sum_{i=2}^{k} \frac{1}{i} + \sum_{i=2}^{n-k+1} \frac{1}{i} = H_k + H_{n-k+1} - 2$$

Subtree sizes

Remark 1.22 (Expected subtree size)

The *expected size* of the *subtree* rooted at the kth internal node is also $H_k + H_{n-k+1} - 1$.



A_x^k

Subtree sizes

Remark 1.22 (Expected subtree size)

The *expected size* of the *subtree* rooted at the kth internal node is also $H_k + H_{n-k+1} - 1$.

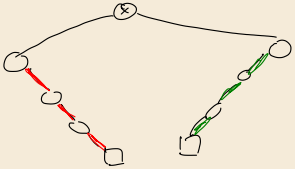
Proof:

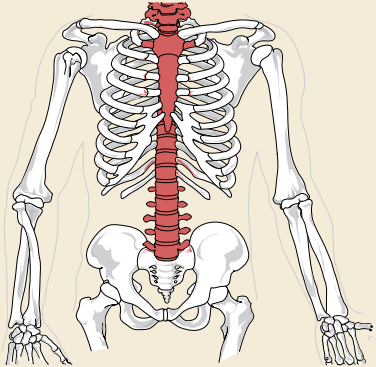
Subtree size of $k = 1 + \sum_{x=1}^{n} A_x^k$

Result follows from similar calculation as above.

$\rightsquigarrow$ Size of secondary data structures to be rebuilt in expectation $2 \ln n$.

Spines of Trees





Lemma 1.23 (Bound on Rotations)
The number of *rotations* to insert or delete a node x in a randomized treap is at most $LS(x) + RS(x)$, where $LS(x)$ and $RS(x)$ are the *lengths of the left resp. right spine* of (the subtree of) x in the treap (after insertion resp. before deletion). ◀

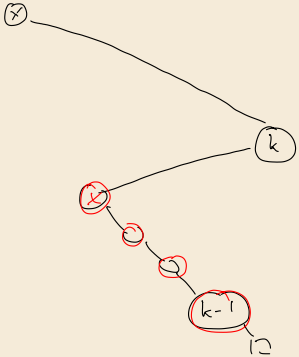
Lemma 1.24 (Expected Spine Lengths)
The expected length of the left and right spine of (the subtree of) the kth internal node (for $k = 1, \ldots, n$) in random BST of n keys are given by

$$\mathbb{E}[LS(k)] = 1 - \frac{1}{k} \qquad \mathbb{E}[RS(k)] = 1 - \frac{1}{n-k+1}$$ ◀

Spines of Trees – Proof

Proof:

$$LS(k) = \sum_{x=1}^{k-1} \left(A_{k-1}^{x} - C_{k-1,k}^{x} \right)$$



1.7 Insertion at Root

Simplerer!

- The details of the implementation of Treaps still need
  - cases distinctions for which rotation to use
  - both code to bubble up nodes (insert) and trickle down (delete)
- We can indeed simplify this further
  (at the expense for more pointer writes)

Simplerer!

- The details of the implementation of Treaps still need
  - cases distinctions for which rotation to use
  - both code to bubble up nodes (insert) and trickle down (delete)
- We can indeed simplify this further
  (at the expense for more pointer writes)
- Idea actually an alternative to standard BST insert / delete
  - Instead of adding a new leaf upon insert,
    we force the new key to **become the root** upon insert
  - $\rightsquigarrow$ need a method to ***split*** a BST into $< x$ and $> x$ for a key x.
    (assume that x not in the tree)
  - if we also have a complementary ***join***,
    deleting an arbitrary node works by joining its children.

Branchless Children

A neat trick makes the following code more compact: store children in 2-element array

```
class Node {
    int key, size = 1;
    Node[] child = new Node[2]; // child[0] is Left, child[1] is Right
}
```

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}
```

For rank-based operations, we store the subtree size of a node

The following helper method is used to keep subtree size up to date

```
void updateSize(Node t) {
    if (t != null) t.size = 1 + size(t.child) + size(t.child);
}
```

Split

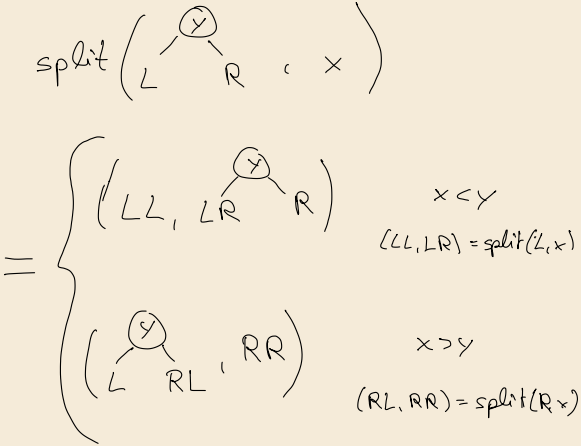
Split BST into two BSTs containing keys $< x$ and $\geq x$ resp.

```
Node[] split(Node t, int x) {
    if (t == null) return new Node[]{null, null};

    int dir = x < t.key ? 0 : 1;
    Node[] res = split(t.child[dir], x);

    // Opposite child receives the remainder of split
    t.child[dir] = res[1 - dir];
    updateSize(t);

    // Construct result dynamically based on dir
    Node[] ans = new Node[2];
    ans[dir] = res[dir];
    ans[1 - dir] = t;
    return ans;
}
```

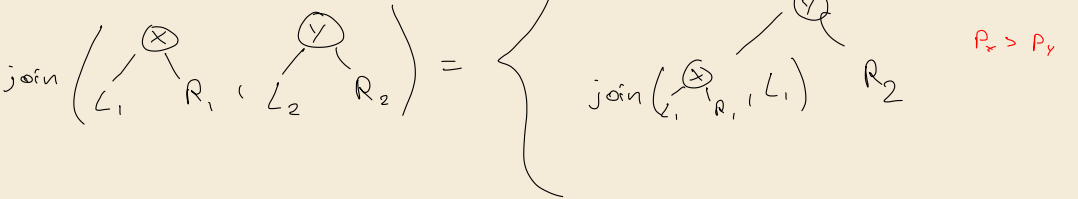


Join

Given two BSTs where all keys in the left are smaller than all keys in the right, merge them into a single BST.

```
Node join(Node left, Node right) {
    if (left == null) return right;
    if (right == null) return left;

    // new root chosen arbitrarily; here larger tree
    int dir = left.size > right.size ? 0 : 1;
    Node root = dir == 0 ? left : right;
    root.child = (dir == 0) ?
        join(root.child[1], right) :
        join(left, root.child[0]);
    updateSize(root);
    return root;
}
```



Insert at Root, Delete

Based on split and merge, insert and delete easy to implement.

```
Node insertAtRoot(Node t, int x) {
    Node[] splits = split(t, x);
    Node root = new Node(x);
    root.child = splits;
    updateSize(root);
    return root;
}

Node delete(Node t, int x) {
    if (t == null) return null;
    if (x == t.key) return join(t.child, t.child);
    int dir = x < t.key ? 0 : 1;
    t.child[dir] = delete(t.child[dir], x);
    updateSize(t);
    return t;
}
```

Tamio Nakajima's Treap Implementation

```cpp
#include <random>
using namespace std;

struct node {
    node *l, *r;
    int prio, key, sum;
};
node nil_node = {&nil_node, &nil_node, 0, 0, 0};
node *nil = &nil_node; // Sentinel node

node *mod_child(node *n0, int id, node *child) {
    node *n = new node;
    *n = *n0;
    (id == 1 ? n->r : n->l) = child;
    n->sum = n->l->sum + n->key + n->r->sum;
    return n;
}

node *join(node *l, node *r) {
    return l == nil ? r : r == nil ? l :
        l->prio > r->prio ?
            mod_child(l, 1, join(l->r, r)) :
            mod_child(r, 0, join(l, r->l));
}

pair<node*,node*> split(node* n, int x) {
    pair<node*, node*> ret = {nil, nil};
    return n == nil ? ret : n->key < x ?
        (ret = split(n->r, x), ret.first =
            mod_child(n, 1, ret.first), ret) :
        (ret = split(n->l, x), ret.second =
            mod_child(n, 0, ret.second), ret);
}

node *mk_node(int x){
    static mt19937 mt(random_device{}());
    int prio = uniform_int_distribution<int>(
        1, 1000000000)(mt);
    return new node {nil, nil, prio, x };
}


node *insert(node *root, int x){
    auto p = split(root, x);
    return join(join(p.first, mk_node(x)), p.second);
}

node *empty_treap(){ return nil; }
```

1.8 Randomized Binary Search Trees

What's wrong with Treaps?

Weaknesses of treaps:

- priorities *fixed once and for all* $\rightsquigarrow$ never recovers from "bad luck"
- have to store *priorities* (at least in a direct implementation), but these are *not helpful* algorithmically.

Randomized Binary Search Trees (RBSTs)

Recall: Key property in random BSTs is that in every subtree of size m, each key value is the root of the subtree with probability $1/m$.

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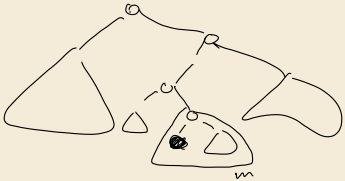
- Store in each node x the size of its subtree $S(x)$.
- **Insert:** Insert x as new leaf and let $y_1, \ldots, y_d$ be the nodes on the path from the root. For each y_j, x should have a $1/S(y_j)$ chance to replace y_j as the subtree root.



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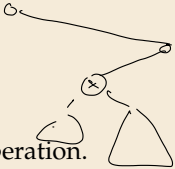
- Store in each node x the size of its subtree $S(x)$.
- **Insert:** Insert x as new leaf and let $y_1, \ldots, y_d$ be the nodes on the path from the root.
  For each y_j, x should have a $1/S(y_j)$ chance to replace y_j as the subtree root.
- **Delete:** After x is gone, one of the remaining $S(x) - 1$ nodes must become subtree root.

  $\rightsquigarrow$ choose one of x's children y and z

  with probabilities $\dfrac{S(y)}{S(y) + S(z)}$ resp. $\dfrac{S(z)}{S(y) + S(z)}$.



Benefits: Tree occasionally rebuilt, subtree sizes useful for rank-based operation.

At each step, flip a biased coin to decide whether x should be subtree root or not

⇝ either call insertAtRoot or recurse.

```
Node insert(Node root, int x) {
    n = root.size;
    r = random.nextInt(n); // uniform int in [0..n)
    if (r == n) // w/p 1/n, x is new root
        return insertAtRoot(root, x);
    if (x < root.key)
        root.left  = insert(root.left,  x);
    else
        root.right = insert(root.right, x);
    updateSize(root);
    return root;
}
```

Insert in RBSTs

Proceed top-down.
At each step, flip a biased coin to decide whether x should be subtree root or not
$\rightsquigarrow$ either call insertAtRoot or recurse.

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    updateSize(root);
    return root;
}
```

Delete in RBSTs

```
Node delete(Node root, int x) {
    if (root == null) return null;
    if      (x < root.key)
        root.left = delete(root.left, x);
    else if (x > root.key)
        root.right = delete(root.right, x);
    else { // must delete root
        root = join(root.left, root.right);
        updateSize(root);
        return root;
    }
}
```

```
Node join(Node left, Node right) {
    if (left == null) return right;
    if (right == null) return left;
    int sl = left.size, sr = right.size;
    r = random.nextInt(sl+sr); // uniform [0..sl+sr)
    int dir = r < sl ? 0 : 1; // 0 w/p sl/(sl+sr)
    Node root = dir == 0 ? left : right;
    root.child = (dir == 0) ?
        join(root.child[1], right) :
        join(left, root.child[0]);
    updateSize(root);
    return root;
}
```

update size correct?

RBST Analysis

Theorem 1.25 (Correctness)

Any sequence of insert and delete operations results in a tree whose shape is that of a *random BST*.

Proof: see paper

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The costs (#visited nodes) of operations in RBSTs on n keys are

(a) same as in random BSTs for **(un)successful search**, i. e., $\sim 2 \ln n$ in expectation and $O(\log n)$ w.h.p.;

(b) **insert** additionally needs $O(1)$ in expectation during insertAtRoot / split, and

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In the *resulting* tree, nodes touched in split are precisely those on the left and right spine of new node x.

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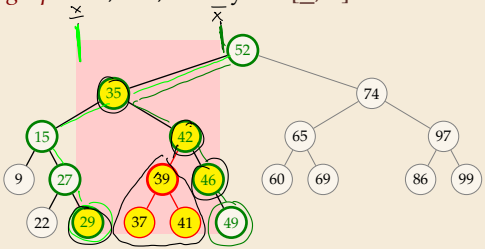
(c) Similar. ■

1.9 Skip Lists

Maybe we don't even want a tree?

Typical application for sorted dictionaries: *(1D) range queries*, i. e., all keys in $[\underline{x}, \overline{x}]$

- BSTs can do that in $O(\log n)$ time
  - search for $\underline{x}$ and $\overline{x}$
  - report nodes on path in range and *inside nodes* (children of path nodes)
  - $\rightsquigarrow$ $O(\log n)$ nodes describe output irrespective of #keys in output



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- inside nodes: nodes between search paths
- outside nodes

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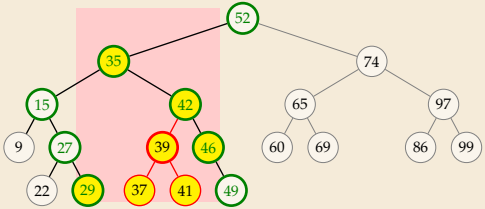

- boundary nodes: nodes in search paths
- inside nodes: nodes between search paths
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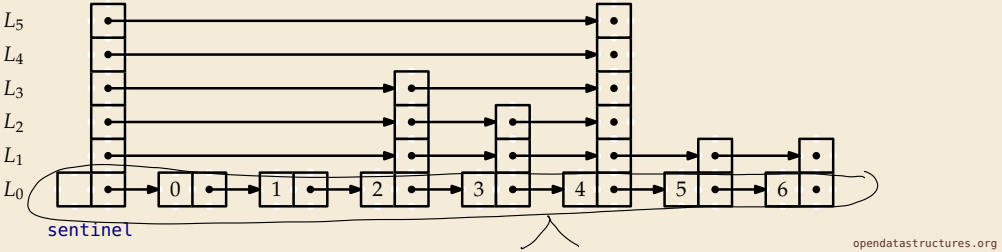
- still, iteration over sorted range rather slow
- needs stack or parent pointers
- a sorted linked list does that faster!

Skiplists

Basic structure of a *skip list*: linked list with shortcuts



insert cheap

once we know location

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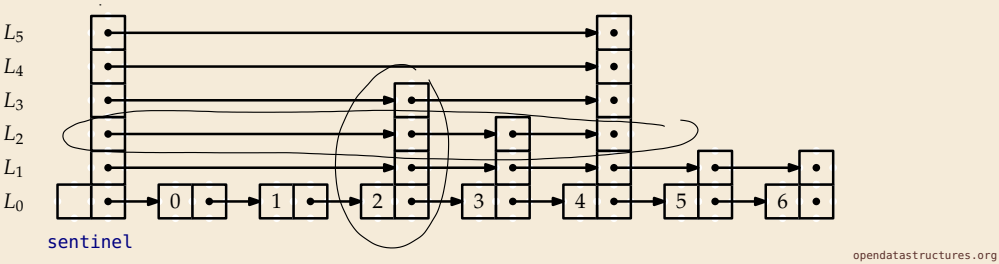

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- each L_{i+1} stores a subsequence of L_i

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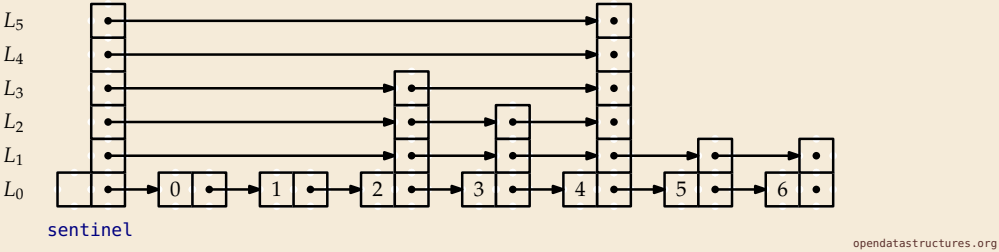

- conceptually sequence of sorted linked lists $L_0, \ldots, L_h$
- each L_{i+1} stores a subsequence of L_i
- obtained by randomized procedure:
  - for each element x in L_i, flip a fair coin $C \stackrel{\mathcal{D}}{=} \mathrm{B}(\frac{1}{2})$
  - if $C = 1$, include x in L_{i+1} (otherwise not)
- eventually stop when next layer empty

Searching in a Skiplist

- head of all lists is sentinel (key $-\infty$)
- each node stores one key and
  a **tower of pointers** `next`$[i]$ for each of its levels i

Searching key x

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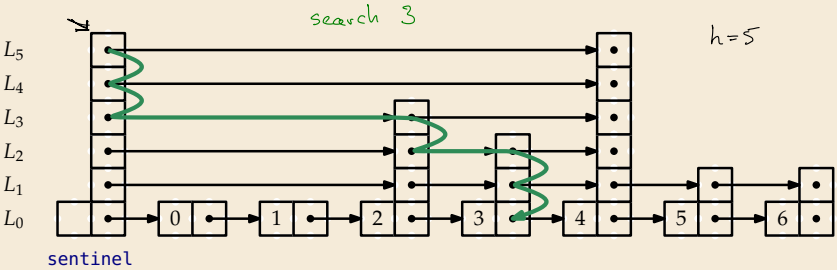
Searching key x

1. Start in top left corner: $\ell := h$, $u :=$ first node of L_h
2. If u.`next`$[\ell]$.`key` $\leq x$ set $u := u$.`next`$[\ell]$ (move right)
   else $\ell := \ell - 1$ (move down)
3. Repeat until $\ell == 0$



Analysis of Skiplist

Recall: #coin tosses until first 1 =: $T \stackrel{\mathcal{D}}{=} \mathrm{Geo}(p)$ with $p = \frac{1}{2}$ (Geometric distribution)

- $\mathbb{P}[T = k] = (1-p)^{k-1} \cdot p$ $\quad k \geq 1$
- $\mathbb{E}[T] = 1/p$ $\quad \mathbb{E}[T] = 2$

$\mathbb{E}[T > k] = (1-p)^{-k}$ $\quad 2^{-k}$

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The expected height $\mathbb{E}[h]$ of a skiplist on n elements is at most $\lg n + 2$. ◀

highest level => h+1 levels

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Theorem 1.29 (Skiplist path lengths)

The expected length of the search path for any node, u, in L_0 is at most $2 \lg n + O(1)$. ◀

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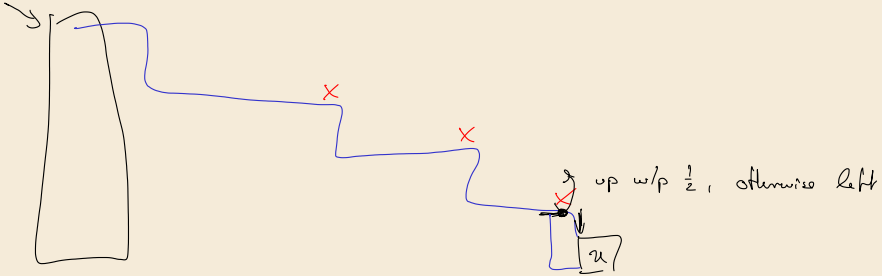

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$S = \sum S_k + h$

S_k = #horizontal steps at level k. $\rightsquigarrow$ Above: $\mathbb{E}[S_k] \leq 1$.

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Total length of path $S = h + S_0 + S_1 + S_2 + \cdots$

$$\mathbb{E}[S] \leq \mathbb{E}[h] + \sum_{k=0}^{\lfloor \lg n \rfloor} \mathbb{E}[S_k] + \sum_{k=\lfloor \lg n \rfloor + 1}^{\infty} \mathbb{E}[S_k]$$

Analysis of Search

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The expected length of the search path for any node, u, in L_0 is at most $2 \lg n + O(1)$. ◀

Proof:
Consider search path *in reverse*.

In each step, if current tower allows to go up, the path does so.
If it cannot go up, it goes left.

How often do we go left before we move up in expectation? $\rightsquigarrow$ coin toss until success!
$\rightsquigarrow$ Expected 1 step left before we go up.

S_k = #horizontal steps at level k. $\rightsquigarrow$ Above: $\mathbb{E}[S_k] \leq 1$.

"$\leq$" since we may run out of nodes in L_k! $\rightsquigarrow$ $\mathbb{E}[S_k] \leq \mathbb{E}[|L_k|] \leq n/2^k$.

Total length of path $S = h + S_0 + S_1 + S_2 + \cdots$

$$\mathbb{E}[S] \leq \mathbb{E}[h] + \sum_{k=0}^{\lfloor \lg n \rfloor} \mathbb{E}[S_k] + \sum_{k=\lfloor \lg n \rfloor + 1}^{\infty} \mathbb{E}[S_k] \leq \mathbb{E}[h] + \sum_{k=0}^{\lfloor \lg n \rfloor} 1 + \sum_{k=\lfloor \lg n \rfloor + 1}^{\infty} \frac{n}{2^k}$$

Analysis of Search

$2 \underline{\ln} n = 1.39 \lg n$

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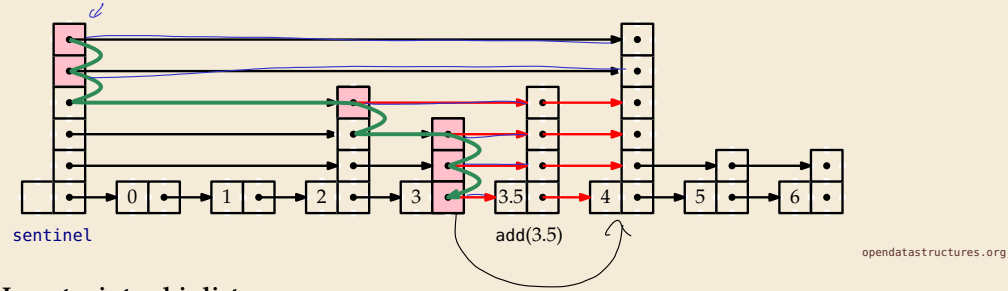
1.10 Operations in Skiplists

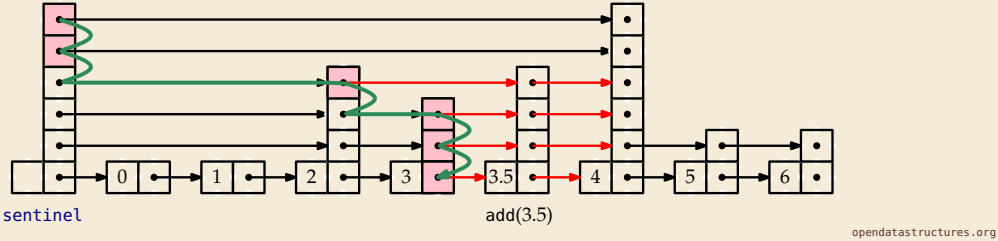
Insert in Skiplists



Insert x into skiplist

1. Search for x, keep "hand" of predecessor nodes on all levels
2. Insert x in a new node u into L_0
3. Add a tower of height $T = \text{Geo}(\frac{1}{2})$ to u
4. Use hand to update predecessors on bottom T levels.

Insert in Skiplists



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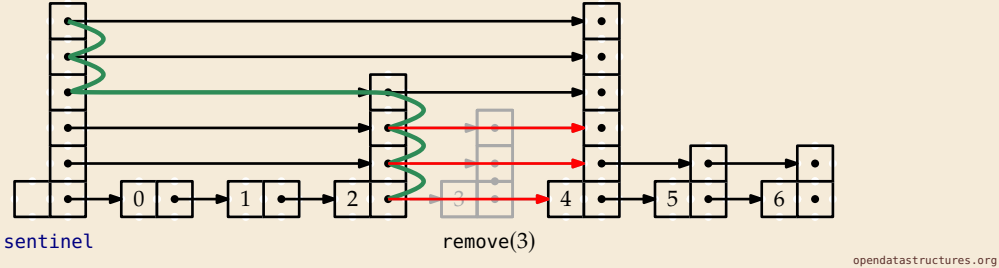
Cost bounded in terms of search path $\rightsquigarrow$ $O(\log n)$ in expectation.

Delete in Skiplists



Delete x from skiplist

1. Search for x, keep "hand" of predecessor nodes on all levels

2. Remove node u containing x from linked lists

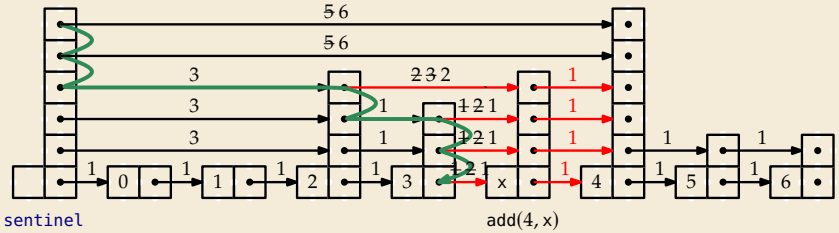
Rank-based operations

Can also support rank-based operations in skiplists.

- Need to know how much we skip!

⇝ Store *length* of pointers

⇝ Can be maintained upon updates (details a bit messy, see *Open Data Structures*)



opendatastructures.org

Skiplists – Discussion

- Extremely simple use of randomness
- Range queries simply follow linked list

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- Extremely simple use of randomness
- Range queries simply follow linked list

- Non-constant size nodes (towers), random space usage
- Worse search costs: $2\,\mathbf{lg}(n)$ vs $2\,\mathbf{ln}(n) \approx 1.39 \lg n$